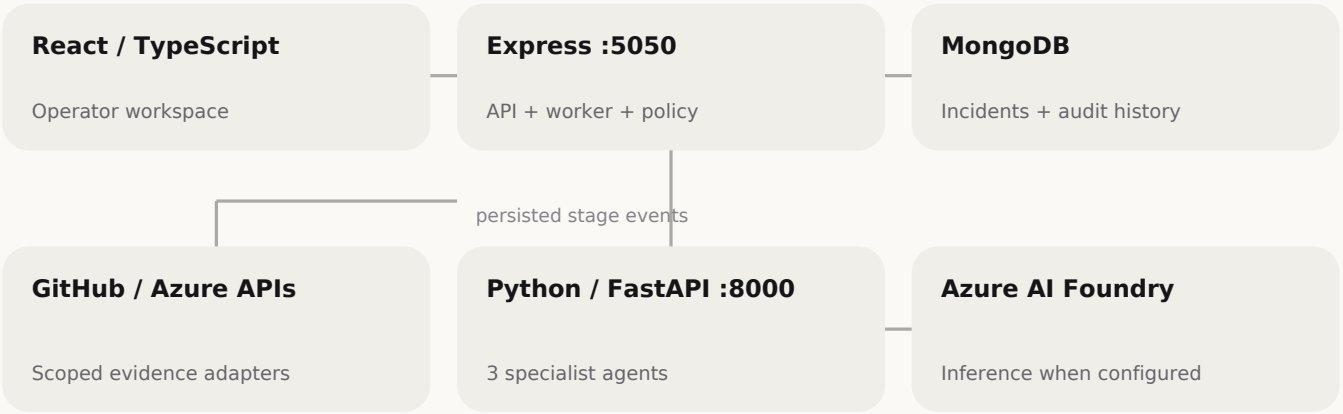
Azure AI engineering & DevOps operations

A full-stack incident-management platform connecting GitHub delivery evidence, Azure diagnostics, three Azure AI Foundry agent stages, durable workflow control and human-authorized remediation.

Service architecture



Canonical incident lifecycle

MongoDB persists the authoritative workflow, run attempts, evidence and audit trail. Version-fenced transitions and idempotent submission protect against stale or repeated actions. Reopening preserves historical records.

Technical design decisions

Execution	Express owns incident workflow and leased worker; Python/FastAPI streams ordered agent stages over NDJSON. Gateway clients consume persisted SSE events with polling fallback.
Cloud evidence	GitHub REST/webhooks and Azure Resource Manager/Monitor adapters collect bounded, source-linked observations. Unavailable telemetry is not reported as healthy.
Persistence	MongoDB/Mongoose: InvestigationRun, IncidentWorkflow, EngineeringSignal, EngineeringEvidence, RemediationProposal and verification records.
Change control	Accepted findings precede proposal review; approved work is recorded externally. Default runtime does not perform Azure/cloud remediation.

Interfaces, safeguards & test evidence

Selected API and event contracts

POST /api/incidents	Authenticated creation with Idempotency-Key; creates or attaches a durable investigation.
POST /api/signals/github	HMAC-SHA256 signed workflow/deployment signal ingestion with provenance and replay deduplication.
POST /api/signals/azure	Authenticated trusted alert-relay ingress; rejects untrusted source claims and disallowed resources.
POST /analyse-stream	Internal FastAPI NDJSON specialist start, result and failure events; actual elapsed time retained.
POST /api/verification/incidents/:id	Azure Monitor baseline/comparison windows must be compatible and complete; result does not automatically resolve the incident.
POST /api/github-changes/:intentId/confirm	Optional, default-off reviewed draft PR with exact content hash, scoped paths, role/approval checks; no merge or deployment.

Security boundaries

Sessions: HttpOnly SameSite=Strict cookies; CSRF token for cookie-authenticated mutations; configured actor/role access. External input: source URL checks, bounded log/patch captures, redaction, allowlisted Azure resources and fixed diagnostic query templates. Approval: model output cannot independently authorize writes, pass verification or close incidents.

Verified local implementation / 2 October 2026

Node tests	53 passed; 1 optional configured-Mongo smoke test skipped; zero failures.
Python tests	11 unittest cases passed.
Browser tests	4 Playwright journeys passed, using real local services/MongoDB and deterministic external provider fixtures.
Frontend	TypeScript/Vite production build and lint passed on Node 24.

External verification boundaries

No production Azure deployment, live Foundry inference/model-quality certification, live provider writes, autonomous remediation or enterprise Entra SSO was established by the final implementation checks. Optional Ollama embeddings require a configured real local model; lexical search is the disclosed fallback.

Source: github.com/keenosmith-del/microsoft-ai-powered-software-engineering-devops | README.md; docs/phase-3/ARCHITECTURE.md; API_REFERENCE.md; SECURITY.md; TEST_RESULTS.md; FINAL_IMPLEMENTATION_RESULTS.md.